

Neural Differential Utility: Endogenous Service Contracts and Certified Dynamic Optimization

Anonymous manuscript for Operations Research

Abstract. A service provider may jointly select operating capacity and a costly contract that changes future cash flows. We study this decision through Neural Differential Utility, retaining a fixed external accounting rule. In a perishable-inventory model, sufficiently responsive contract premiums yield optimal stock and contract choices that increase with demand stress and the inherited contract. We derive an exact upper-envelope algorithm, smooth neural critics with uniform error bounds, and a quadratic contract-discretization bound without global concavity. Expanded-action dynamic programming represents the model exactly; the economic gain is relative to restricted contract choices, not to unrestricted stochastic control. Exact rational calculations under a previously recorded protocol improve net reward and physical cost over a frozen contract while reducing the demand fill rate. A physical-cost oracle remains better on its own objective. For continuous reconfiguration, we establish a boundary-complete value and policy certificate and a scheme-specific refinement bound. Direct value gradients avoid singular volatility inversion, but an identical-data least-squares experiment shows parameterization parity. Exact finite-model results, analytic diffusion certificates, and historical neural diagnostics are reported separately.

Keywords: stochastic control; inventory; service contracts; endogenous adjustment; value gradients; numerical certification.

Subject classifications: Inventory/production: service contracts; Dynamic programming/optimal control: stochastic control; Analysis of algorithms: approximation guarantees.

Area of review: Stochastic Models.

1 Introduction

A provider that promises a higher service level can earn a higher premium, incur a larger liability when demand is not served, and pay to maintain or revise the commitment. The operating decision and the contract decision are therefore connected. Stock determines exposure to liability; the contract changes the marginal value of stock; and an inherited commitment makes rapid adjustment costly. The relevant question is not whether a controller may redefine an unfavorable score. It is how a provider should jointly choose a physical action and an externally priced, persistent service commitment.

We develop this operational interpretation of Neural Differential Utility. The term denotes joint control of physical activity and a costly internal or contractual state. The external accounting rule remains fixed throughout optimization and evaluation. The framework includes fixed-contract decision rules, and it is itself an ordinary stochastic-control problem when both decisions are admitted to the action set. This exact representation is useful: it supplies a stringent reference computation and makes the source of any improvement unambiguous.

Our first contribution is a structural result for a perishable-inventory model. When incremental premium growth compensates for the incremental liability associated with demand stress, optimal stock and contract choices are monotone in both the demand regime and the inherited contract. The condition is imposed on cash-flow primitives and is verified for the recorded experiment. It is not inferred from the sign of a controllable penalty. Indeed, a provider allowed merely to lower its own shortage penalty would generally reduce that penalty when exposure rises. The distinction separates an economic service commitment from a change in the scoring rule.

Our second contribution is computational. For each contract, stock can be eliminated by a newsvendor calculation. The remaining value function, after adding back the quadratic adjustment term, is an upper envelope of affine functions of the inherited contract. This representation gives an exact threshold algorithm on finite contract grids. More importantly, the same envelope property yields a quadratic contract-discretization error bound even when the reduced optimization problem is not globally concave. We use the bound to enclose the value of the continuous-contract problem, rather than treating a finer numerical grid as an exact reference.

Our third contribution concerns economic interpretation and evidence. We derive an accounting identity linking changes in physical cost to changes in optimized net reward and contract transfers. It applies to the actual optimizing policy, not to a service parameter selected afterward to match a desired critical fractile. We execute all five methods in a previously recorded protocol with exact rational dynamic programming and independent recomputation. The joint policy improves net reward and physical cost over the fixed contract, but it serves a smaller fraction of demand. A state-aware physical-cost oracle achieves a lower physical cost. These comparisons identify a trade-off rather than universal operational dominance.

The continuous-time part addresses a related computational difficulty. A contract or internal state can evolve nearly deterministically even though its shadow price matters for control. The volatility of the value process then provides a poorly conditioned observation of that shadow price. A direct value-gradient representation avoids the inversion, provided it remains compatible with a scalar value function. We prove a stopped-domain certificate with separate terminal and lateral-boundary errors and with both

value comparisons needed for a policy-loss bound. A manufactured controlled diffusion supplies analytic certificates and a separate space–time–action refinement calculation. These results do not turn a sampled residual from a large neural experiment into a uniform certificate.

Joint pricing and replenishment are established stochastic-control decisions; for example, Federgruen and Heching (1999) analyze demand-dependent pricing with inventory replenishment and backlogging. Here demand is exogenous, stock perishes, and the service contract changes premium and shortage liability while carrying an adjustment state. We do not claim a new general class of joint control. The useful structure is the premium–liability condition, the inherited-contract envelope, and the resulting computable error budget.

The structural argument builds on lattice optimization (Topkis, 1978); the contribution is the premium–liability condition and its dynamic propagation in the contract model. The computational analysis uses the geometry of the Bellman maximization, rather than an assumption that every inner problem is concave. Our continuous-time representation retains the value-process and gradient relationship used in forward–backward methods (Ma et al., 1994; Han et al., 2018). Our quantitative grid statement is proved for the displayed scheme and solution; it does not infer a numerical rate solely from qualitative convergence conditions.

2 An externally priced inventory contract

There are periods $t = 0, \dots, T - 1$. The observed state is (z, q) , where z is a demand regime in a finite ordered set $\mathcal{Z}$ and $q \in [0, 1]$ is the inherited contract. Conditional demand D_z is nonnegative with finite first moment. A decision consists of stock $S \in \mathcal{S} \subset [0, \bar{S}]$ and contract $\theta \in \Theta \subset [0, 1]$. Both feasible sets are fixed compact chains; finite grids are allowed. Unused stock perishes and unmet demand is lost. Thus no inventory or backlog is carried. The next regime has exogenous transition matrix P , and the next inherited contract equals the chosen contract.

Write the expected overage and shortage as

$$O_z(S) = \mathbb{E}(S - D_z)^+, \quad L_z(S) = \mathbb{E}(D_z - S)^+, \quad C_z(S) = cS + hO_z(S) + pL_z(S). \quad (1)$$

Here $c, h, p \geq 0$ are externally fixed physical-cost coefficients. The provider receives premium $r_z\theta$, pays incremental liability $\kappa\theta L_z(S)$, pays maintenance $m\theta^2$, and pays adjustment $\lambda(\theta - q)^2$, where $\kappa, m, \lambda > 0$. Expected one-period net reward is

$$g(z, q, S, \theta) = r_z\theta - C_z(S) - \kappa\theta L_z(S) - m\theta^2 - \lambda(\theta - q)^2. \quad (2)$$

The expectation in this expression is over demand, not over the decision. Premium rates, liability coefficients, and physical-cost weights cannot be controlled. The contract is a real cash-flow commitment. Its level affects both reward and the future state; it is not an estimator of a preference parameter.

For discount $\beta \in (0, 1]$ and zero terminal reward, the Bellman equation is

$$V_t(z, q) = \max_{S, \theta} \left\{ g(z, q, S, \theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta) \right\}, \quad V_T = 0. \quad (3)$$

Finite actions make existence immediate. For compact intervals, bounded demand moments, continuity of the stage reward, and induction give a continuous value and an attained maximum. We select the greatest optimal pair when the lattice conditions below hold.

2.1 Exact representation and comparator classes

Proposition 2.1 (Expanded actions and the fixed-contract restriction). *The model in (3) is exactly the Markov decision process with state (z, q) , action (S, θ) , reward (2), and transition $P_{zz'} \mathbf{1}_{\{q'=\theta\}}$. Its value is unchanged by flattening the product action set into one action index. A frozen-contract problem is the restriction $\theta = \bar{\theta}$; its value cannot exceed the joint value when the same initial state and external reward are used. A dummy label that changes neither reward nor the next state does not enlarge the attainable value.*

Proof. A policy in either representation assigns the same pair to each observed history, so it induces the same joint law of states, actions, and rewards. Equivalently, the two Bellman maxima enumerate the same quantities. The frozen problem optimizes over a subset of these policies. Duplicating a maximand leaves a maximum unchanged. $\square$

Optimizing over a genuine extra decision can nevertheless produce a reduced operator that is not affine in the continuation value. For fixed physical action, write

$$\mathcal{B}(v) = \sup_{\theta} \{b_{\theta} + \beta \langle P_{\theta}, v \rangle\}. \quad (4)$$

If this envelope has two different uniquely exposed slopes on an open convex set of continuation values, it cannot equal one fixed affine reward/kernel primitive there. This is a statement about eliminating a decision while retaining one primitive, not a separation from the expanded-action problem. The proof and the precise exposed-slope condition appear in Electronic Companion (EC) Section EC.1.

Nor does support-envelope geometry exclude entropic control. For $\rho > 0$, a finite full-support reference distribution P , and any bounded vector v ,

$$\frac{1}{\rho} \log \mathbb{E}_P e^{\rho \beta v} = \sup_{Q \in \Delta} \left\{ \beta \mathbb{E}_Q v - \frac{1}{\rho} \text{KL}(Q \| P) \right\}. \quad (5)$$

The optimizer is proportional to $P \exp(\rho \beta v)$. Subtracting the objective at a particular Q from the left side gives $\rho^{-1} \text{KL}(Q \| Q^*) \geq 0$. A sufficiently rich kernel-control family therefore represents this entropic operator exactly. The inventory model uses its specific contractual family; the general framework does not claim non-equivalence to all risk-sensitive or robust-control models.

3 Structure, exact reduction, and approximation

Assumption 3.1 (Premium–liability compatibility). Demand is stochastically nondecreasing in z , the regime transition is stochastically monotone, and for every $z_2 \geq z_1$ and feasible S ,

$$r_{z_2} - r_{z_1} \geq \kappa \{L_{z_2}(S) - L_{z_1}(S)\}. \quad (6)$$

All other cost coefficients and the product action set are independent of z .

For a translated demand family $D_z = D_0 + z$, shortage is 1-Lipschitz in the translation. Thus $r_{z_2} - r_{z_1} \geq \kappa(z_2 - z_1)$ is a sufficient primitive condition. It allows shortage exposure to rise; premium growth must compensate for the incremental liability attached to a larger contract.

Theorem 3.2 (Monotone stock and contract). *Under Assumption 3.1, V_t is supermodular in (z, q) for every t . The set of maximizing pairs in (3) is a compact sublattice and has a greatest element. Both components of this greatest optimizer, $S_t^*(z, q)$ and $\theta_t^*(z, q)$, are nondecreasing in z and in q .*

Proof. We verify increasing differences for the full maximand, including the continuation value. In the action pair (S, θ) , the only cross term is $-\kappa\theta L_z(S)$, which has increasing differences because L_z decreases in S . In (θ, q) , the cross difference from adjustment is $2\lambda \Delta\theta \Delta q \geq 0$.

For fixed θ , $-C_z(S) - \kappa\theta L_z(S)$ has increasing differences in (S, z) . To see this without a density assumption, its stock marginal at continuity points is

$$-c - hF_z(S) + (p + \kappa\theta)(1 - F_z(S));$$

stochastic demand ordering makes this marginal nondecreasing in z . Integrating between two stock levels proves the required finite-difference inequality, including at atoms. The stage cross difference in (θ, z) is

$$\Delta\theta [\Delta r - \kappa\Delta L(S)] \geq 0$$

by (6). All remaining cross pairs involving q or z have zero stage cross difference.

Suppose V_{t+1} is supermodular in (z', q') . For $\theta_2 \geq \theta_1$, the function $z' \mapsto V_{t+1}(z', \theta_2) - V_{t+1}(z', \theta_1)$ is nondecreasing. Stochastic monotonicity of P therefore makes its conditional expectation nondecreasing in z . The continuation term has increasing differences in (z, θ) and is independent of S, q . Consequently the full maximand is supermodular on the product lattice (S, θ, z, q) .

Maximizing a supermodular function over a fixed compact action lattice preserves supermodularity in the remaining coordinates: apply the lattice inequality to maximizing pairs at two parameter states, and use their meet and join as feasible pairs at the meet and join of those states. The same inequality makes the argmax a sublattice and orders its greatest elements when the parameters increase. This is the lattice optimization argument of Topkis (1978). Since $V_T = 0$ is supermodular, backward induction completes the proof. $\square$

The result concerns the state variables of the stated model. It does not assert a backlog comparative static in a model that has no carried backlog. It also does not follow for an internal penalty-only objective. With terminal value zero, $p(\theta) = \theta$, $q = 1/2$, and fixed stock, that alternative objective has optimizer

$$\arg \max_{0 \leq \theta \leq 1} \{-\theta L - \lambda(\theta - 1/2)^2\} = \text{proj}_{[0,1]} \left(\frac{1}{2} - \frac{L}{2\lambda} \right), \quad (7)$$

which decreases with exposure. External premium income in (2), and specifically (6), changes this conclusion.

3.1 Stock elimination and the inherited-contract envelope

For a fixed contract, stock solves a newsvendor problem:

$$S_z(\theta) \in \arg \min_{S \in \mathcal{S}} \{cS + hO_z(S) + (p + \kappa\theta)L_z(S)\}. \quad (8)$$

On a stock interval with a continuous strictly increasing demand distribution, an interior optimum has critical fractile

$$F_z(S_z(\theta)) = \frac{p + \kappa\theta - c}{h + p + \kappa\theta}. \quad (9)$$

Boundary solutions apply when the right side is outside the attainable distribution range. For atoms or a stock grid, (8) itself, with the greatest-minimizer convention, is the definition; no continuous quantile formula is silently substituted for grid optimization.

Set $U_z(\theta) = \max_S \{-C_z(S) - \kappa\theta L_z(S)\}$ and

$$b_t(z, \theta) = r_z\theta - (m + \lambda)\theta^2 + U_z(\theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta). \quad (10)$$

Then

$$V_t(z, q) = -\lambda q^2 + \max_{\theta \in \Theta} \{b_t(z, \theta) + 2\lambda\theta q\}. \quad (11)$$

Theorem 3.3 (Exact envelope reduction). *For any demand family and any regime transition, $q \mapsto V_t(z, q) + \lambda q^2$ is convex. If Θ contains N_θ ordered points, this function is a convex piecewise-affine envelope with ordered slopes $2\lambda\theta$. Its active line selects a greatest optimal contract, and (8) supplies the corresponding stock. After stock costs and continuation expectations are computed, the envelope can be constructed in $O(N_\theta)$ arithmetic operations from the ordered slopes and queried at N_q ordered inherited-contract states in $O(N_\theta + N_q)$ operations. No global concavity of b_t is required.*

Proof. Equation (11) is obtained by expanding $-\lambda(\theta - q)^2$. A supremum of affine functions is convex. For finite Θ , insert lines in increasing slope order. Intersections with the current last line must increase along an upper hull; otherwise the last line has an empty interval of optimality and is removed. Each line is inserted and removed at most once. At an intersection choose the larger slope, which selects the greater contract. A single forward pass through ordered queries locates all active intervals. Stock elimination is exact because the continuation and adjustment terms do not depend on stock. $\square$

The operation count is an arithmetic count, not a bit-complexity statement for arbitrarily long rational numbers. A direct factored enumeration already reduces the expensive product search: on the recorded grid, it uses 2,904 rather than 37,752 Bellman action comparisons, plus 429 one-time stock-stage computations. The hull construction further avoids enumerating every contract at every inherited state. The executed hull and factored enumeration agree with exhaustive rational optimization at all 264 nonterminal states.

3.2 A quadratic bound without global concavity

Let V now allow every contract in $[0, 1]$, retaining the same stock set. Let $V^{(n)}$ restrict all contract actions to $\Theta_n = \{0, 1/n, \dots, 1\}$ but permit any initial $q \in [0, 1]$. After the first action, the inherited state lies on the grid. Define $H_r = \sum_{k=0}^{r-1} \beta^k$, with $H_0 = 0$.

Theorem 3.4 (Continuous-contract value enclosure). *With the primitives of (2), for $r = T - t \geq 1$,*

$$0 \leq V_t(z, q) - V_t^{(n)}(z, q) \leq \frac{(m + \lambda)H_r + \beta\lambda H_{r-1}}{4n^2}. \quad (12)$$

This is also a performance-loss bound for the exact grid-optimal policy when it is implemented in the continuous-contract problem from (z, q) . It is uniform in z, q and does not require differentiability of demand distributions, global concavity, or a numerical smoothness estimate.

Proof. First consider $f(x) = G(x) - Ax^2 + \ell x$ on $[0, 1]$, where G is finite and convex and $A > 0$. If a maximum x^* lies at an endpoint, it belongs to every grid. If it is interior, take any subgradient s of G there. The supporting-line inequality gives

$$f(x) \geq f(x^*) + (s - 2Ax^* + \ell)(x - x^*) - A(x - x^*)^2.$$

An interior maximum forces the linear coefficient to vanish: otherwise a sufficiently small displacement with its sign would improve the objective. Therefore the nearest grid point loses at most $A/(4n^2)$.

For the Bellman maximization after stock elimination, U_z is convex as a supremum of affine functions of θ . Theorem 3.3, applied to continuous actions, says that $V_{t+1}(z', \theta) + \lambda\theta^2$ is convex. Consequently its contract objective has the displayed form with $A = m + \lambda + \beta\lambda$ whenever $t < T - 1$. At the final period, $A = m + \lambda$ because $V_T = 0$. Restricting just the current maximization thus loses at most the corresponding $A/(4n^2)$.

Let $E_t = \sup_{z,q} (V_t - V_t^{(n)})$. Monotonicity and the β -Lipschitz property of the Bellman operator give

$$E_t \leq \frac{m + \lambda + \beta\lambda \mathbf{1}_{\{t < T-1\}}}{4n^2} + \beta E_{t+1}, \quad E_T = 0.$$

Summing proves (12). A grid policy uses admissible continuous actions and has exactly the grid-model reward and transition law, so its continuous-model value is $V^{(n)}$. $\square$

The theorem bounds contract discretization, not physical-stock discretization or an inventory diffusion limit. It avoids interpreting a small difference between two numerical grids as a proof. In the recorded instance, the 160-interval result encloses the continuous-contract initial value in $[-4.99278573, -4.99260152]$; the exact rational endpoints are supplied with the code.

3.3 Constructive smooth neural critics

The exact finite-grid envelope also supplies a smooth neural approximation with a deterministic error budget. This construction uses computed envelope weights rather than statistical training. It provides a certified neural

representation of the contract model without making a claim about the optimization of a large network.

Corollary 3.5 (Smooth value-gradient representation). *For a finite contract grid, let $V_t^{(n)}$ be the exact grid value, extended to every inherited $q \in [0, 1]$ by (11). For each (t, z) write its convex part as*

$$V_t^{(n)}(z, q) + \lambda q^2 = a + bq + \sum_j d_j(q - k_j)^+, \quad d_j \geq 0, \quad \sum_j d_j \leq 2\lambda.$$

For $\tau > 0$, replace each hinge by

$$h_\tau(x) = \frac{(x + \tau)_+^3 - 2x_+^3 + (x - \tau)_+^3}{6\tau^2}$$

and denote the resulting critic by $w_{t,\tau}$; keep $w_{T,\tau} = 0$. It is a quadratic skip connection plus a finite cubic-ReLU network, is C^2 in q , and satisfies

$$0 \leq w_{t,\tau}(z, q) - V_t^{(n)}(z, q) \leq \lambda\tau/3. \quad (13)$$

If A_τ exactly maximizes the one-step grid objective with continuation $w_{t+1,\tau}$, then, for $r = T - t$,

$$0 \leq V_t^{(n)}(z, q) - V_t^{A_\tau}(z, q) \leq \beta(\lambda\tau/3)H_{r-1}. \quad (14)$$

Combining this bound with (12) bounds loss relative to continuous contract actions. The derivative $\partial_q w_{t,\tau}$ is a compatible shadow price, not an independently fitted vector field.

Proof. The hinge representation follows from the ordered slopes of the upper envelope. Its total slope increase is at most 2λ , the width of the slope interval. The cubic finite difference equals the convolution of x^+ with the symmetric triangular density $(\tau - |y|)_+/\tau^2$. Convexity gives $h_\tau(x) \geq x^+$, and direct integration shows $h_\tau(x) - x^+ \leq \tau/6$, with equality at $x = 0$. This proves (13). Each cubic-ReLU term is C^2 .

The exact and approximate one-step action values differ by a number in $[0, \beta\lambda\tau/3]$. Thus maximizing the latter loses at most $\beta\lambda\tau/3$ in the former, rather than twice that amount, because the approximation is one-sided. The final action uses the exact terminal continuation and has zero loss. Backward policy evaluation gives (14). The gradient statement follows from the scalar construction. $\square$

4 Net reward, physical cost, and service

For a policy π , let C^π be its expected discounted physical cost and let K^π be its expected discounted premium minus liability, maintenance, and adjustment. Its net reward is $W^\pi = K^\pi - C^\pi$. All these quantities use the same external coefficients. Let J denote a net-reward-optimal joint policy, F a frozen-contract policy optimized for that reward, and O a physical-cost-optimal policy with the same information and admissible stock actions.

Theorem 4.1 (Accounting and an optimizer-based physical-cost bound). *For these actual policies,*

$$C^F - C^J = (W^J - W^F) - (K^J - K^F), \quad (15)$$

$$0 \leq C^J - C^O \leq K^J - K^O. \quad (16)$$

Thus J improves physical cost over F exactly when its optimized net-reward gain exceeds the increase in net contract transfers. Physical-cost dominance over O cannot follow from net-reward optimality. A physical oracle may have multiple contract implementations; (16) holds for any specified feasible implementation.

Proof. Subtract $W = K - C$ for the two policies to obtain (15). Physical optimality gives $C^O \leq C^J$. Net-reward optimality gives $K^J - C^J \geq K^O - C^O$, yielding the other inequality. Neither step inserts a preferred contract into the optimizing policy. $\square$

The bound can be strengthened by choosing the net-transfer-maximizing implementation within the physical-optimal policy set. Our primary experiment does not change the previously recorded oracle implementation for this purpose: the greatest-contract tie convention selects contract one. The physical lower bound is independent of that convention, whereas the oracle's reported net cash flow is not.

A positive physical gap can also be stable without assuming that all reoptimized policies remain unchanged. Suppose the same two calibrated policies are deployed under a perturbed exogenous one-step law whose total-variation distance is at most η uniformly in regime, and physical stage cost lies in $[0, \bar{C}]$. Coupling exogenous histories gives

$$|(C^F - C^J)_{\text{new}} - (C^F - C^J)_{\text{old}}| \leq 2\bar{C}\eta \sum_{t=0}^{T-1} \beta^t(t+1). \quad (17)$$

Before a coupling failure the two copies of each fixed policy choose identical contracts and stock. The chance of failure by period t is at most $(t+1)\eta$; summation proves the bound. This statement concerns deployment of the calibrated policies, not arbitrary policy reselection at a tie. The traditional critical-fractile mismatch argument is retained separately in EC.2; it is not used to attribute physical improvement to an unproved endogenous selector.

5 Continuous reconfiguration and boundary-complete computation

The continuous-time counterpart allows a physical state c and an internal or contractual state u to evolve under $\alpha = (a, \theta)$. On a domain K , consider

$$dX_s = b(X_s, \alpha_s) ds + \Sigma_\varepsilon(X_s, \alpha_s) dB_s, \quad X = (c, u), \quad \Sigma_\varepsilon = \begin{bmatrix} \sigma_c & 0 \\ 0 & \sqrt{2\varepsilon}I \end{bmatrix} \quad (18)$$

in the block-diagonal special case. A controlled internal ordinary differential equation is obtained at $\varepsilon = 0$. A neural differential parameterization can specify its drift; it does not alter the external reward rule. This continuous model is not claimed to be a diffusion limit of the perishable finite experiment.

5.1 Value gradients, volatility, and conditioning

For a smooth value function v , define column vectors $P = \nabla v$ and $Z = \Sigma_\varepsilon^\top P$. Along an optimal trajectory, before stopping,

$$dY_s = -(r(X_s, \alpha_s) - \rho Y_s) ds + Z_s^\top dB_s, \quad Y_s = v(s, X_s), \quad (19)$$

with the appropriate terminal or exit payoff. When the diffusion is control-independent, the maximizing drift-reward expression is $P^\top b + r$. For a controlled diffusion, the Hessian term must also be optimized. We never infer a feedback law from Z alone when the map $P \mapsto \Sigma^\top P$ has a nontrivial kernel. This is the value-process representation underlying the forward-backward viewpoint (Ma et al., 1994); deep BSDE approximations provide one numerical parameterization (Han et al., 2018).

Lemma 5.1 (Singular inversion and compatible direct gradients). *In the preference block of (18), $Z_u = \sqrt{2\varepsilon}P_u$. If $\widehat{Z}_u = Z_u + \eta_Z$ is inverted, then*

$$\mathbb{E} \left\| \widehat{P}_u - P_u \right\|^2 = \frac{\mathbb{E} \|\eta_Z\|^2}{2\varepsilon}. \quad (20)$$

A direct error $\widehat{P}_u = P_u + \eta_P$ has mean-square error $\mathbb{E} \|\eta_P\|^2$ without this inversion factor. At $\varepsilon = 0$, Z_u contains no information about P_u . Taking $P_\phi = \nabla v_\phi$ enforces gradient compatibility, but by itself proves neither an HJB residual bound nor policy optimality.

Proof. Subtract the two volatility identities and divide by $\sqrt{2\varepsilon}$. Squaring and taking expectations gives (20). The zero-noise assertion follows directly from the linear map. A scalar critic has a conservative gradient wherever it is differentiable; optimality requires the separate residual and action conditions below. $\square$

Equal-scale additive target errors are an assumption of the conditioning comparison, not a consequence of learning from identical observations. In the recorded least-squares experiment, the direct and volatility parameterizations span precisely the same fitted functions. Their recovered gradients agree to floating-point precision. The observed parity is consistent with (20): naturally induced volatility error can shrink with the noise level, so a fixed target-error floor need not apply.

5.2 A stopped Dirichlet problem

Fix $D = [0, T] \times \bar{K}$ and the stopping time $\tau = T \wedge \inf\{s \geq t : X_s \notin K\}$. On exit before T , pay $\psi(\tau, X_\tau)$; at T , pay $G(X_T)$. The value is

$$V^*(t, x) = \sup_{\alpha} \mathbb{E}_{t,x} \left[\int_t^\tau e^{-\rho(s-t)} r(X_s, \alpha_s) ds + e^{-\rho(\tau-t)} \Phi(\tau, X_\tau) \right], \quad (21)$$

where $\Phi = \psi$ on lateral faces and $\Phi = G$ on the terminal face, with compatible corner data. Assume bounded data, compact controls, well-posed stopped dynamics, admissible arbitrarily accurate Hamiltonian selectors, and the dynamic programming principle and comparison for this Dirichlet problem. Uniform ellipticity and a sufficiently regular bounded domain are one sufficient setting; the theorem below also applies when the

stated probabilistic assumptions hold for a degenerate model. Write

$$H^\alpha[w] = r^\alpha + b^\alpha \cdot \nabla w + \frac{1}{2} \text{tr}(\Sigma^\alpha \Sigma^{\alpha^\top} D^2 w), \quad F[w] = w_t + \sup_\alpha H^\alpha[w] - \rho w.$$

The problem is $F[V^*] = 0$ in the interior, $V^* = \psi$ on the lateral boundary, and $V^*(T, \cdot) = G$. Reflected and state-constrained problems use different boundary operators and are not implicitly substituted for this one.

Theorem 5.2 (Absolute value and policy certificates). *Let $w \in C^{1,2}(D)$ satisfy the integrability conditions for stopped Itô's formula. Let A be a measurable admissible actor. Suppose*

$$\begin{aligned} |F[w]| &\leq \delta_H, & 0 \leq \sup_\alpha H^\alpha[w] - H^A[w] &\leq \delta_A \quad \text{in the interior,} \\ |w(T, x) - G(x)| &\leq \delta_T, & |w(s, x) - \psi(s, x)| &\leq \delta_\partial \quad \text{on lateral faces.} \end{aligned}$$

Put $\delta_B = \max(\delta_T, \delta_\partial)$ and $H_\rho(s) = (1 - e^{-\rho s})/\rho$ for $\rho > 0$, with $H_0(s) = s$. Then

$$|w(t, x) - V^*(t, x)| \leq \delta_B + H_\rho(T - t)\delta_H, \quad (22)$$

$$0 \leq V^*(t, x) - V^A(t, x) \leq 2\delta_B + H_\rho(T - t)(2\delta_H + \delta_A). \quad (23)$$

If instead the certified residual is $|w_t + H^A[w] - \rho w| \leq \tilde{\delta}_H$, a direct bound is

$$V^* - V^A \leq 2\delta_B + H_\rho(T - t)(2\tilde{\delta}_H + \delta_A). \quad (24)$$

A sequence whose relevant absolute defects vanish therefore has uniformly convergent values and asymptotically optimal actors on this stopped problem.

Proof. For any admissible control, stopped Itô's formula gives

$$J^\alpha - w(t, x) = \mathbb{E} \left[e^{-\rho(\tau-t)} (\Phi - w)(\tau, X_\tau) + \int_t^\tau e^{-\rho(s-t)} (w_t + H^\alpha[w] - \rho w)(s, X_s) ds \right].$$

The integrand is at most δ_H for every control. Approximate measurable maximizers make it at least $-\delta_H$ up to an arbitrarily small error. The boundary contribution has absolute value at most δ_B , proving (22). For actor A , the integrand lies between $-\delta_H - \delta_A$ and δ_H , so $w - V^A \leq \delta_B + H_\rho(T - t)(\delta_H + \delta_A)$. Adding this comparison to the upper bound on $V^* - w$ proves (23). In particular, both boundary comparisons must be counted.

For (24), the actor integrand has absolute value at most $\tilde{\delta}_H$, while every competing integrand is at most $\tilde{\delta}_H + \delta_A$. Applying the same identity once to a competitor and once to A gives the displayed result. Bounded stopping horizons justify the use of $H_\rho(T - t)$. $\square$

A sampled root-mean-square residual is not a bound on δ_H . A deterministic validation cover can supply one only with an independently justified modulus: if $F[w]$ is L_R -Lipschitz and the cover has fill distance h , then $\delta_H \leq \max_{\text{cover}} |F[w]| + L_R h$. Terminal and lateral defects require their own covers and moduli. The

action gap needs a separate enclosure over all admissible actions. EC.3 retains these verification details and a counterexample to omitting the lateral boundary.

5.3 A constructive instance and a scheme-specific bound

Consider $T = 1$, $K = (-1, 1)^2$, controls $a, \theta \in [-1, 1]$, no discount, and

$$\mathcal{L}^{a,\theta} = a\partial_c + \frac{\theta}{4}\partial_u + 0.045\partial_{cc} + 0.08\partial_{uu}, \quad (25)$$

$$r(c, u, a, \theta) = -a^2 - \frac{5}{4}\theta^2 - \frac{1}{4}c^2 - \frac{1}{80}u^2 - 0.125. \quad (26)$$

Pay $(c^2 + u^2)/2$ on the entire parabolic boundary. Then $V^* = (c^2 + u^2)/2$: the maximizing controls are $a = c/2$ and $\theta = u/10$, their maximized terms cancel the state costs, and the diffusion contribution is 0.125.

For $e \in [0, 1/16]$, define $w_e = V^* + e(1-t)(1-c^2)(1-u^2)$. Its exact greedy actor is $a_e = (w_e)_c/2$, $\theta_e = (w_e)_u/10$, which remains inside the control box. Direct expansion gives

$$\delta_T = \delta_\partial = \delta_A = 0, \quad \delta_H \leq 2.30e + 1.05e^2. \quad (27)$$

Indeed, the time and diffusion terms contribute at most $(1+0.09+0.16)e$, the linear Hamiltonian perturbations at most $(1+0.05)e$, and the squared perturbations at most $(1+0.05)e^2$. Thus (22)–(23) give an analytic uniform certificate sequence as $e \downarrow 0$. This is a manufactured controlled diffusion, not a trained inventory critic.

We also solve a genuinely refined discrete problem using a locally consistent Markov-chain construction of the type developed by Kushner and Dupuis (2001). With spatial step h , forward/backward differences D^+, D^- , central second differences D_h^2 , and $b^+ = \max(b, 0)$, $b^- = \min(b, 0)$, define

$$\begin{aligned} L_h^{a,\theta} v &= a^+ D_c^+ v + a^- D_c^- v + \frac{1}{4}(\theta^+ D_u^+ v + \theta^- D_u^- v) + 0.045 D_{cc,h}^2 v + 0.08 D_{uu,h}^2 v, \\ T_h v &= \max_{a,\theta \in A_k} \{v + \Delta t (L_h^{a,\theta} v + r^{a,\theta})\}, \end{aligned}$$

where $A_k = \{-1, -1 + 2/k, \dots, 1\}$. Boundary nodes receive the prescribed payoff at every time. If

$$\Delta t \left(0.25h^{-2} + 1.25h^{-1} \right) \leq 1, \quad (28)$$

all transition weights are nonnegative and sum to one.

Proposition 5.3 (Quantitative refinement for the displayed scheme). *For (25)–(26), exact terminal and lateral data, and (28), the discrete backward solution v_h satisfies, at every state–time node,*

$$|v_h - V^*| \leq (1-t) \left(\frac{5}{8}h + \frac{9}{4k^2} \right). \quad (29)$$

This is a grid-value bound for this scheme and solution, not a certificate for an arbitrary interpolated neural actor.

Proof. Central differences of V^* are exact. The upwind first derivative contributes an additional $h|a|/2$ and $h|\theta|/8$ to the generator, together at most $5h/8$. Nearest grid controls are within $1/k$ of the continuous maximizers. Completing squares in the two Hamiltonians bounds their joint quantization loss by $(1+5/4)/k^2$. The exact solution is time-independent, so no temporal truncation term remains for this instance. Hence the one-step defect of V^* is at most $\Delta t(5h/8 + 9/(4k^2))$. Nonnegative transition weights make T_h nonexpansive in the sup norm. Backward induction from the exact parabolic data adds one local defect per step and proves (29). $\square$

Unlike a fixed state grid with more actions, this construction refines state, time, and actions together. A more general smooth solution would require the corresponding spatial and temporal truncation terms; merely naming monotonicity, stability, and consistency would not establish (29).

6 Computational study

6.1 Protocol, computation, and economic comparisons

The recorded protocol fixes $T = 8$, $\beta = 0.97$, $z \in \{0, 1, 2\}$, and $D_z = z + D_0$, where $D_0 \in \{0, 1, 2, 3\}$ has probabilities $(0.1, 0.3, 0.4, 0.2)$. The transition is

$$P = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.15 & 0.70 & 0.15 \\ 0.05 & 0.25 & 0.70 \end{pmatrix}.$$

The coefficients are $c = 0.3$, $h = 0.6$, $p = 0.65$, $\kappa = 1.2$, $m = 1.5$, $\lambda = 0.6$, and $r_z = 0.6 + 1.4z$. Stock is $0, 0.5, \dots, 6$, contract is $0, 0.1, \dots, 1$, and the initial state is $(1, 0.5)$. Demand translation makes $\Delta L \leq \Delta z$, while $\Delta r = 1.4\Delta z > 1.2\Delta z$. The transition rows are stochastically ordered. Thus the structural theorem applies to exactly the model computed.

All five methods use unrestricted state-aware stock tables. Joint contract and expanded-action MDP enumerate all 143 action pairs. Frozen contract fixes the economic contract at 0.5 and has 13 distinct stock actions. The capacity-matched gate enumerates 11 dummy labels for each stock choice but leaves the economic contract and next inherited state at 0.5; it still has only 13 distinct economic actions. The physical oracle minimizes physical cost. Under the recorded greatest-pair convention it selects contract one on contract ties; only its physical-cost comparison is invariant to this implementation convention.

Table 1 reports exact discounted expectations. Joint and expanded-action values and policies agree at every nonterminal state, as do frozen and dummy-gate values and policies. The independent Fraction implementation checks 1,320 state–method records; the monotonicity check covers 416 adjacent state pairs. The hull critic agrees at all 264 states. Exact Bellman, terminal, greedy, and accounting defects are zero for the finite model.

Net reward improves from -5.947058 to -5.002311 , a gain of 0.944747 . Physical cost falls by 0.213204 , from 9.587378 to 9.374174 . Nevertheless, the fill rate falls from 0.925926 to 0.882109 . The physical oracle costs 8.506095 , below both contract policies. These results have a direct economic reading: the joint contract

changes the cash-flow value of carrying stock, and it economizes on physical cost partly by accepting more lost demand. Improved welfare is not a synonym for higher service.

The full accounting decomposition appears in Table 2. For example, the joint policy earns more premium but also pays more liability, maintenance, and adjustment. Equation 15 accounts for all these changes. No historical total-cost field is relabeled as theta-free physical cost.

The simulation uses 64 batches of 4,096 episodes with common exogenous paths across methods and the recorded seed sequence. Two-sided paired Student intervals use 63 degrees of freedom and quantify simulation error only. The paired joint-minus-frozen net-reward difference is 0.943729 with interval $[0.939821, 0.947636]$; the physical-cost difference is -0.212112 with interval $[-0.214933, -0.209291]$. Exact expectations, rather than Monte Carlo significance, establish the finite-model ordering. Batch-level outputs also retain undiscounted service counts.

6.2 Contract refinement, conditioning, and diffusion certificates

Table 3 reports the recorded nested contract grids with 10, 20, 40, 80, and 160 intervals. Each row solves the same stock-grid model exactly. Theorem 3.4 provides the continuous-contract enclosure, with error bounds decreasing from 0.047158 to 0.00018421. These are analytic error budgets, not thresholds fitted to observed grid differences. A first-order Lipschitz budget is retained in the machine-readable output for comparison.

The conditioning experiment uses 64 batches of 2,048 identical observations for each parameterization. For u uniform on $[-1, 1]$ and standard normal ξ , the observed payoff is $(u + \sqrt{2\varepsilon}\xi)^2/2$. We fit $ku^2/2 + b$ and $\zeta u^2/(2\sqrt{2\varepsilon}) + b$ by unregularized least squares. The resulting shadow prices are ku and $\zeta u/\sqrt{2\varepsilon}$. Table 4 shows matching mean-square errors; recovered slopes agree to floating-point precision in both the original execution and the independent replay. The volatility design becomes less well-conditioned at the smallest noise level, but the exact least-squares fit does not display a shadow-price accuracy advantage for the direct parameterization. No artificial target-noise floor is added to create one.

We additionally compile the smooth critics of Corollary 3.5 at $\tau = 2^{-m}$ for $m = 4, 8, 12, 16, 20$. They use at most 27 cubic-ReLU units per state–time critic. Exact rational evaluation checks every one of the 264 nonterminal grid states at each level. All five induced policies coincide with the exact grid optimizer at these states, giving zero observed grid-policy loss. The proved uniform policy budget, which also covers initial inherited contracts between grid points, decreases from 0.0776069 to 1.18419×10^{-6} . EC.4 reports the weights, finite-model interpretation, and complete audit locations. This is constructive approximation and validation, not a newly trained neural solver.

Table 5 instantiates (27) at the five recorded perturbation levels. It reports separate terminal and lateral defects and the absolute policy-loss bound. The bound decreases to approximately 4.38690×10^{-6} . The state–time–action study in Table 6 is an additional deterministic scheme check, separate from the recorded analytic sequence. It refines 25 to 1,089 spatial nodes, 4 to 94 time steps, and 9 to 65 actions per coordinate. The observed all-node error decreases from 0.0445124 to 0.00742613, within the proved scheme-specific bounds. Floating-point observed errors are diagnostics; the bounds follow from the analytic truncation calculation and nonexpansive propagation.

6.3 Evidence retained from earlier versions

The complete earlier manuscript, proofs, figures, numerical outputs, release archive, and reproducibility checks remain available in the historical snapshot. EC.5 identifies their relationship to the present paper. Historical large-neural inventory residual ratios and greedy-gap summaries are not reclassified as uniform certificates: they lack the absolute parabolic-boundary and interior enclosures required by Theorem 5.2. The finite model, the manufactured diffusion, and those sampled neural diagnostics are different mathematical objects. A successful exact calculation on one does not supply missing hypotheses for another.

7 Operational implications and reproducibility

A costly contract has operational content when its cash flows are priced externally and its persistence affects future decisions. The premium–liability condition explains when stress raises the chosen commitment rather than inducing the provider to lower an internal score. The envelope representation turns the inherited contract into an ordered threshold decision and supplies an approximation guarantee without global concavity. These results are fully compatible with ordinary expanded-action stochastic control.

The numerical evidence supports a specific, reproducible conclusion: joint contracting improves net reward and physical cost over the recorded fixed-contract policy, while sacrificing fill rate and remaining above the physical-cost lower bound. For continuous reconfiguration, direct value gradients address a conditioning issue, whereas residual, greedy, terminal, and lateral defects determine certification. These roles should be kept distinct in algorithm design and in operational interpretation.

Code and data availability. The revision package contains executable integer and independently written rational dynamic programs, all policy and accounting records, paired simulation and conditioning outputs, analytic and grid-refinement calculations, and a point-by-point response. The entry point is `revisions/or-r4-20260920/execute.py`; generated evidence is in its `results` directory. The frozen protocol is retained at `revisions/or-r2-20260920/protocol.json`. No empirical data are needed for the new experiments. Earlier data and source artifacts are preserved with their original interpretation in the historical snapshot, not silently pooled with the new results.

Acknowledgments

Author-identifying acknowledgments are omitted for anonymous review.

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Table 1: Exact discounted operating outcomes

Method	Net reward	Physical cost	Fill rate	Actions
Joint contract	-5.002311	9.374174	0.882109	143
Expanded-action MDP	-5.002311	9.374174	0.882109	143
Frozen contract	-5.947058	9.587378	0.925926	13
Dummy gate	-5.947058	9.587378	0.925926	13
Physical oracle	-11.972030	8.506095	0.703704	143

All expectations are exact rational calculations; displayed decimals are rounded. Actions count distinct economic choices. The physical oracle minimizes physical cost, not the convention-dependent net reward shown here.

Table 2: Contract cash-flow decomposition

Method	Premium	Liability	Maintenance	Adjustment
Joint contract	9.889658	1.001859	4.308149	0.207786
Expanded-action MDP	9.889658	1.001859	4.308149	0.207786
Frozen contract	7.208555	0.865027	2.703208	0.000000
Dummy gate	7.208555	0.865027	2.703208	0.000000
Physical oracle	14.417109	6.920212	10.812832	0.150000

Net reward equals premium minus physical cost, liability, maintenance, and adjustment. The oracle uses contract one under the recorded tie convention.

Table 3: Exact grid values and continuous-contract enclosures

Intervals	Grid value	Loss bound	Value upper bound
10	-5.00231055	0.04715774	-4.95515281
20	-4.99648554	0.01178944	-4.98469611
40	-4.99331402	0.00294736	-4.99036666
80	-4.99288305	0.00073684	-4.99214621
160	-4.99278573	0.00018421	-4.99260152

Theorem 3.4 bounds contract discretization for the unchanged stock set. The lower endpoint is the grid value; the upper endpoint adds the proved bound.

Table 4: Identical-data least-squares conditioning experiment

ε	Direct P MSE	P MSE via Z	Direct condition	Z condition
0.1	0.00131353	0.00131353	6.901	3.461
0.01	0.00010902	0.00010902	6.901	2.980
0.001	0.00001052	0.00001052	6.901	7.653
0.0001	0.00000104	0.00000104	6.901	23.719

Means over 64 batches, each with 2,048 pairs. Integrated shadow-price MSE uses uniform test states on $[-1, 1]$. Condition numbers are spectral design-matrix condition numbers.

Table 5: Absolute certificates for the stopped diffusion

m	δ_H bound	δ_T	δ_∂	δ_A	Policy-loss bound
4	0.147851563	0	0	0	0.295703125
8	0.009000397	0	0	0	0.018000793
12	0.000561586	0	0	0	0.001123172
16	0.000035095	0	0	0	0.000070191
20	0.000002193	0	0	0	0.000004387

$e = 2^{-m}$. The entire parabolic boundary is exact, and the actor is analytically greedy. These are manufactured analytic critics, not fitted neural inventory models.

Table 6: Joint space–time–action refinement

Spatial nodes	Time steps	Actions per axis	Node error	Proved bound
25	4	9	0.04451240	0.34765625
81	10	17	0.02567135	0.16503906
289	29	33	0.01404964	0.08032227
1089	94	65	0.00742613	0.03961182

Observed errors are maximized over every computed state–time node. The bounds are supplied by Proposition 5.3; they do not certify interpolation or a continuous-state actor.